

Mandatory exercises 1

1. Solve:

$$-2w + 2x + 2y - 8z = 14$$

$$3w - 3x + y + 4z = -5$$

$$2w - 2x - 4y + 12z = -22$$

$$\begin{bmatrix} -2 & 2 & 2 & -8 & 14 \\ 3 & -3 & 1 & 4 & -5 \\ 2 & -2 & -4 & 12 & -22 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 \cdot \frac{1}{2}} \begin{bmatrix} -1 & 1 & 1 & -4 & 7 \\ 3 & -3 & 1 & 4 & -5 \\ 2 & -2 & -4 & 12 & -22 \end{bmatrix} \begin{array}{l} R_2 \leftarrow R_2 + 3 \cdot R_1 \\ R_3 \leftarrow R_3 + 2 \cdot R_1 \end{array}$$

$$\begin{bmatrix} -1 & 1 & 1 & -4 & 7 \\ 0 & 0 & 4 & -8 & 16 \\ 0 & 0 & 2 & 4 & -8 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 \cdot \frac{1}{2}} \begin{bmatrix} -1 & 1 & 1 & -4 & 7 \\ 0 & 0 & 2 & 4 & 8 \\ 0 & 0 & 2 & 4 & -8 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2}$$

$$\begin{bmatrix} -1 & 1 & 1 & -4 & 7 \\ 0 & 0 & 2 & 4 & 8 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\begin{array}{l} R_1 \leftarrow R_1 \cdot (-1) \\ R_2 \leftarrow R_2 \cdot (\frac{1}{2}) \end{array}} \begin{bmatrix} 1 & -1 & 1 & 4 & -7 \\ 0 & 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 + R_2}$$

$$\begin{bmatrix} 1 & -1 & 0 & 2 & -3 \\ 0 & 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} P \\ F \\ P \\ P \\ F \end{array}$$

① $1w - 1x + 2z = -3$
 $\hookrightarrow \boxed{w = -3 + x - 2z}$

② $1y - 2z = 4$
 $\hookrightarrow \boxed{y = 4 + 2z}$

③ $x, z \rightarrow \text{free}$
 $\hookrightarrow \boxed{x, z \in \mathbb{R}}$

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$$24\text{l (Chem A)} + 32\text{l (chem B)} + 52\text{l (chem C)} = \text{kill}$$

	Parts of Chem A	Parts Chem B	Parts of Chem C
Spray X:	1x	2x	2x
Spray Y:	0y	0y	1y
Spray Z:	$\frac{1}{3}z$	$\frac{1}{3}z$	$\frac{1}{3}z$
	Sum = 24l	Sum = 32l	Sum = 52l

$$\begin{cases} 1x + 0y + \frac{1}{3}z = 24 \\ 2x + 0y + \frac{1}{3}z = 32 \\ 2x + 1y + \frac{1}{3}z = 52 \end{cases}$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{3} & 24 \\ 2 & 0 & \frac{1}{3} & 32 \\ 2 & 1 & \frac{1}{3} & 52 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 + (-1 \cdot R_1)} \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{3} & 24 \\ 1 & 0 & 0 & 8 \\ 2 & 1 & \frac{1}{3} & 52 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 + (-1 \cdot R_1)}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{3} & 24 \\ 1 & 0 & 0 & 8 \\ 1 & 1 & 0 & 28 \end{array} \right] \Rightarrow \begin{cases} 8 + \frac{1}{3}z = 24 \\ \frac{1}{3}z = 16 \quad \times 3 \\ z = 48 \end{cases}$$

$$\rightarrow 8 + y = 28 \\ y = 20$$

$\rightarrow$ 8l of spray X, 20l of Spray Y, 48l of Spray Z

3.4.4) find Adjoint and Calculate A^{-1}

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \\ 2 & 2 & 2 \end{bmatrix}$$

1.) Cofactors using Minors

$$(a \cdot d - b \cdot c)$$

$$M_{11} = \begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix} = 1 \cdot 2 - (-1) \cdot 2 = 4 \rightarrow C_{11} = 4$$

$$M_{12} = \begin{bmatrix} 0 & -1 \\ 2 & 2 \end{bmatrix} = 0 + 2 = 2 \rightarrow C_{12} = M_{12} \cdot (-1) = C_{12} = -2$$

$$M_{13} = -2 \rightarrow C_{13} = -2$$

$$M_{21} = -2 \rightarrow C_{21} = 2$$

$$M_{22} = -4 \rightarrow C_{22} = -4$$

$$M_{23} = -2 \rightarrow C_{23} = 2$$

$$M_{31} = -5 \rightarrow C_{31} = -5$$

$$M_{32} = -1 \rightarrow C_{32} = 1$$

$$M_{33} = 1 \rightarrow C_{33} = 1$$

$$|A| = a_{11} \cdot C_{11} + a_{21} \cdot C_{21} + a_{31} \cdot C_{31}$$

$$= 1 \cdot 4 + 0 \cdot 2 + 2 \cdot (-5)$$

$$4 + 0 - 10$$

$$|A| = -6$$

Cofactor matrix A

Cofactor matrix $A^T = \text{adj}(A)$

$$\begin{bmatrix} 4 & -2 & -2 \\ 2 & -4 & 2 \\ -5 & 1 & 1 \end{bmatrix} \xrightarrow{\text{transform}} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & \dots & \dots \\ a_{31} & \dots & \dots \end{bmatrix} \rightarrow \begin{bmatrix} 4 & 2 & -5 \\ -2 & -4 & 1 \\ -2 & 2 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj}(A) = \frac{1}{-6} \cdot \begin{bmatrix} 4 & 2 & -5 \\ -2 & -4 & 1 \\ -2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{2}{3} & -\frac{1}{3} & \frac{5}{6} \\ \frac{1}{3} & \frac{2}{3} & -\frac{1}{6} \\ \frac{1}{3} & -\frac{1}{3} & -\frac{1}{6} \end{bmatrix}$$

$\rightarrow A^{-1}$